

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Full Stack development MERN Practical

Practical 5

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Class: TYIT	Batch: 1
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AIM :- React Routing and Data Handling

A) Install and configure React Router in the React application.

CODE :-

(run in terminal):- npm install react-router-dom

OUTPUT :-

```
PS C:\Users\mvluc\Desktop\Ritesh(Mern)\prac_5> npm install react-router-dom
up to date, audited 140 packages in 2s
32 packages are looking for funding
  run `npm fund` for details
found 0 vulnerabilities
```

B) Create multiple pages and implement navigation between them.

CODE :-

About.jsx :-

```
function About(){
  return (
    <div>
      <h1>About page</h1>
      <p>This page contains information about our application Ritesh Chaubey</p>
    </div>
  )
}
export default About;
```

Home.jsx :-

```
function Home(){
  return(
    <div>
      <h1>Home Page</h1>
      <p>Welcome to my React Application</p>
    </div>
  )
}
```

```

}
export default Home;
Contact.jsx:-
function Contact(){
return(
<div>
<h1>Contact Page</h1>
<p>Email:Reze@example.com</p>
</div>
)
}

```

```

export default Contact;

```

App.jsx:-

```

import { BrowserRouter, Routes, Route, Link } from "react-router-dom";
import Home from "./pages/Home";
import About from "./pages/About";
import Contact from "./pages/Contact";
function App() {
return (
<BrowserRouter>
<h1>My React Application</h1>
<nav>
<Link to="/">Home</Link>
<br/>
<Link to="/about">About</Link>
<br/>
<Link to="/contact">Contact</Link>
</nav>
<hr />
<Routes>
<Route path="/" element={<Home />} />
<Route path="/about" element={<About />} />
<Route path="/contact" element={<Contact />} />
</Routes>
</BrowserRouter>
);
}
export default App;

```

OUTPUT :-

My React Application

[Home](#)
[About](#)
[Contact](#)

Home Page

Welcome to my React Application

My React Application

[Home](#)
[About](#)
[Contact](#)

About page

This page contains information about our application Ritesh Chaubey

My React Application

[Home](#)
[About](#)
[Contact](#)

Contact Page

Email:Reze@example.com

C) Use Link and Routes to move between pages.

CODE :-

App.jsx :-

```
import { BrowserRouter, Routes, Route, Link } from "react-router-dom";
import Dashboard from "./pages/Dashboard";
import Statistics from "./pages/Statistics";
import Jobs from "./pages/Jobs";
import Profile from "./pages/Profile";
function App() {
  return (
    <BrowserRouter>
      <div>
        <nav>
          <Link to="/">Dashboard</Link> |{" "}
          <Link to="/statistics">Statistics</Link> |{" "}
          <Link to="/jobs">Jobs</Link> |{" "}
          <Link to="/profile">Profile</Link>
        </nav>
        <hr />
        <Routes>
          <Route path="/" element={<Dashboard />} />
          <Route path="/statistics" element={<Statistics />} />
          <Route path="/jobs" element={<Jobs />} />
          <Route path="/profile" element={<Profile />} />
        </Routes>
      </div>
    </BrowserRouter>
  );
}
export default App;
```

Dashboard.jsx :-

```
function Dashboard(){
  return(
    <div>
      <h1>Placement Dashboard</h1>
```

```

        <p>Welcome to Student Placement Portal.</p>
        <h2>Placement Summary</h2>
        <p>Total Companies: 25</p>
        <p>Students Placed: 120</p>
        <p>Highest Package: 12LPA</p>
    </div>
)
}
export default Dashboard;

```

Profile.jsx :-

```

function Profile(){
    return(
        <div>
            <h1>Student Profile</h1>
            <hr />
            <p>Name: Himani Malankar</p>
            <p>CGPA: 8.95</p>
            <p>Skills: Data Science , Data Analytics</p>
        </div>
    )
}
export default Profile;

```

Jobs.jsx :-

```

function Jobs(){
    return(
        <div>
            <h1>Job Opportunities</h1>
            <hr />
            <h2>Software Developer</h2>
            <p>Company: ABC Technologies</p>
            <p>Packages: ₹6LPA</p>
            <hr />
            <h2>Web Developer</h2>
            <p>Company: XYZ Technologies</p>
            <p>Packages: ₹5LPA</p>
            <hr />
            <h2>Data Scientist</h2>
            <p>Company: Entrata Technologies</p>
            <p>Packages: ₹15LPA</p>
            <hr />
            <h2>Data Analyst</h2>
            <p>Company: Accenture Technologies</p>
            <p>Packages: ₹12LPA</p>
        </div>
    )
}
export default Jobs;

```

Statistics.jsx

```

function Statistics(){
    return(
        <div>
            <h1>Placement Statistics</h1>
            <hr />
            <p>Students Registered: 200</p>
            <p>Student Placed: 120</p>
            <p>Placement Percentage: 60%</p>
        </div>
    )
}

```

```

    <p>Average Package: ₹5.5LPA</p>
  </div>
)
}
export default Statistics;

```

OUTPUT :-

Dashboard Statistics Jobs Profile	Dashboard Statistics Jobs Profile
Placement Dashboard Welcome to Student Placement Portal. Placement Summary Total Companies: 25 Students Placed: 120 Highest Package: 12LPA	Placement Statistics Students Registered: 200 Student Placed: 120 Placement Percentage: 60% Average Package: ₹5.5LPA

Dashboard Statistics Jobs Profile
Job Opportunities
Software Developer Company: ABC Technologies Packages: ₹6LPA
Web Developer Company: XYZ Technologies Packages: ₹5LPA
Data Scientist Company: Entrata Technologies Packages: ₹15LPA
Data Analyst Company: Accenture Technologies Packages: ₹12LPA

Dashboard Statistics Jobs Profile
Student Profile
Name: Saqlain Khan CGPA: 8.95 Skills: Data Science , Data Analytics

D) Retrieve data from a public API via Axios.

Setup Axios library

(run in terminal):- npm install axios

CODE :-

```
import axios from "axios";
```

```

import { useEffect , useState } from "react";
function App() {
  const[products , setProducts]=useState([]);
  useEffect(()=> {axios.get("https://dummyjson.com/products")
    .then((response)=>{
      setProducts(response.data.products);
    });
  },[]);
  return(
    <div>
      <h1>Products List</h1>
      <hr/>
      {products.map((product)=>
        <div key ={product.id}>
          <h2>{product.title}</h2>
          <p>Price: Rs. {product.price}</p>
        </div>
      )}
    </div>
  );
}
export default App;

```

OUTPUT :-

Products List
Essence Mascara Lash Princess Price: Rs.9.99 Eyeshadow Palette with Mirror Price: Rs.19.99 Powder Canister Price: Rs.14.99 Red Lipstick Price: Rs.12.99 Red Nail Polish Price: Rs.8.99 Calvin Klein CK One Price: Rs.49.99 Chanel Coco Noir Eau De Price: Rs.129.99 Dior J'adore Price: Rs.89.99 Dolce Shine Eau de Price: Rs.69.99 Gucci Bloom Eau de Price: Rs.79.99 Annibale Colombo Bed

E) Render the collected information within lists or tables.

CODE :-

```

import axios from "axios";
import { useEffect, useState } from "react";
function App() {
  const [products, setProducts] = useState([]);
  useEffect(() => {
    axios
      .get("https://dummyjson.com/products")
      .then((response) => {
        setProducts(response.data.products);
      });
  });
}

```

```

    })
    .catch((error) => {
      console.error("Error fetching products:", error);
    });
  }, []);
return (
  <div>
    <h1>Products List</h1>
    <hr />
    <table border="1">
      <thead>
        <tr>
          <th>ID</th>
          <th>Product</th>
          <th>Price</th>
          <th>Category</th>
        </tr>
      </thead>
      <tbody>
        {products.map((product) => (
          <tr key={product.id}>
            <td>{product.id}</td>
            <td>{product.title}</td>
            <td>{product.price}</td>
            <td>{product.category}</td>
          </tr>
        ))}
      </tbody>
    </table>
  </div>
);
}
export default App;
OUTPUT :-

```

Products List

ID	Product	Price	Category
1	Essence Mascara Lash Princess	9.99	beauty
2	Eyeshadow Palette with Mirror	19.99	beauty
3	Powder Canister	14.99	beauty
4	Red Lipstick	12.99	beauty
5	Red Nail Polish	8.99	beauty
6	Calvin Klein CK One	49.99	fragrances
7	Chanel Coco Noir Eau De	129.99	fragrances
8	Dior J'adore	89.99	fragrances

F) Apply filtering or search capabilities to the visible dataset.
CODE :-

```

import axios from "axios";
import { useEffect, useState } from "react";
function App() {
  const [products, setProducts] = useState([]);
  const [search, setSearch] = useState("");
  useEffect(() => {
    axios.get("https://dummyjson.com/products")
      .then(res => setProducts(res.data.products));
  }, []);
  const filtered = products.filter(p =>
    p.title.toLowerCase().includes(search.toLowerCase())
  );
  return (
    <div>
      <h1>Products List</h1>
      <input
        placeholder="Search..."
        onChange={e => setSearch(e.target.value)}
      />
      <table border="1">
        <thead>
          <tr>
            <th>ID</th>
            <th>Product</th>
            <th>Price</th>
            <th>Category</th>
          </tr>
        </thead>
        <tbody>
          {filtered.map(p => (
            <tr key={p.id}>
              <td>{p.id}</td>
              <td>{p.title}</td>
              <td>{p.price}</td>
              <td>{p.category}</td>
            </tr>
          ))}
        </tbody>
      </table>
    </div>
  );
}
export default App;

```

OUTPUT :-

Products List

ID	Product	Price	Category
1	Essence Mascara Lash Princess	9.99	beauty
2	Eyeshadow Palette with Mirror	19.99	beauty
3	Powder Canister	14.99	beauty
4	Red Lipstick	12.99	beauty
5	Red Nail Polish	8.99	beauty
6	Calvin Klein CK One	49.99	fragrances
7	Chanel Coco Noir Eau De	129.99	fragrances
8	Dior J'adore	89.99	fragrances
9	Dolce Shine Eau de	69.99	fragrances

Practical 4(E):

Create a Movie Watchlist React App

Required fields:

- **Title of movie**
- **Category**
- **Score**

CODE :-

```
import React, { useState } from "react";
function App() {
  const [movies, setMovies] = useState([
    {
      id: 1,
      name: "Inception",
      genre: "Sci-Fi",
      rating: 8.8
    },
    {
      id: 2,
      name: "The Dark Knight",
      genre: "Action",
      rating: 9.0
    },
    {
      id: 3,
      name: "Interstellar",
      genre: "Sci-Fi",
      rating: 8.7
    },
    {
      id: 4,
      name: "3 Idiots",
      genre: "Comedy/Drama",
```

```

        rating: 8.4
      }
    ]);
    return (
      <div className="container">
        <h1>Movie Watchlist</h1>

        <div className="movie-list">
          {movies.map((movie) => (
            <div className="movie-card" key={movie.id}>
              <h2>{movie.name}</h2>
              <p>
                <strong>Genre:</strong> {movie.genre}
              </p>
              <p>
                <strong>Rating:</strong> {movie.rating}
              </p>
            </div>
          ))}
        </div>
      </div>
    );
  }
  export default App;
  OUTPUT :-

```

Movie Watchlist

Inception

Genre:Sci-Fi

Rating:8.8

The Dark Knight

Genre:Action

Rating:9

Interstellar

Genre:Sci-Fi

Rating:8.7

3 Idiots

Genre:Comedy/Drama

Rating:8.4